

Mahek Vanjani

Gujarat, India | +91 9512080717 | vanjanimahek654@gmail.com | linkedin.com/in/mahek-vanjani |
github.com/MahekVanjani611 | Portfolio

EDUCATION

Indian Institute of Technology Jodhpur

Bachelor of Technology in Electrical Engineering; GPA: 8.74/10

Jodhpur, Rajasthan

2022 - 2026

St. Anselm's Sr. Sec. School

HSC-12th; Percentage: 96.6/100

Abu Road, Rajasthan

2021 - 2022

WORK EXPERIENCE

Neural Web

2026

Data Science Intern

Remote

- Delivered AI-powered healthcare solutions using Python and FastAPI, translating client requirements into scalable data-driven applications
- Built automation workflows including Text-to-SQL systems and presentation-generation pipelines to improve data accessibility and reporting efficiency
- Analyzed structured healthcare datasets involving CPT and ICD codes to support clinical decision-making and operational insights

NVIDIA

Summer 2025

Software Engineer Intern

On-site

- Improved Linux GPU driver performance by identifying and resolving NVBug-related bottlenecks, contributing to a 3% FPS increase across AAA gaming workloads
- Optimized a C++ 32,000-node linked-list implementation by redesigning it as an unrolled list (block size 16), reducing cache misses by 94% and lowering traversal latency from 1.6 ms to 0.1 ms
- Performed root-cause analysis of a complex multithreading race condition, strengthening driver stability and reliability for 1,000+ users in developer builds
- Leveraged performance traces, logs, and system-level analysis to investigate runtime issues and improve overall software reliability

PROJECTS

AI Research Agent | Python, LangGraph, ChromaDB, Mem0, MCP

Preview

- Built an agentic research assistant using a RAG pipeline (ChromaDB vector store + custom document loader) to ground LLM responses in user-uploaded documents and reduce hallucination on domain queries
- Designed a LangGraph-based multi-step reasoning workflow (`research_graph.py`) to decompose research queries into retrieval, synthesis, and answer-generation stages
- Integrated Mem0 for persistent cross-session memory, enabling the agent to recall prior context and user preferences across chat sessions
- Implemented an MCP (Model Context Protocol) server to expose agent tools to external LLM clients, enabling standardized interoperable tool-calling
- Added multi-user authentication and session management, supporting isolated per-user chat history and secure login **GitHub Repository**

Set Transformer with Attention-Aware Dynamic Neural Network for Adaptive PSL Replication |

- Built an attention-based Set Transformer to replicate Programmable Scheme Logic (PSL) in protection IEDs using Digital Data Block (DDB) signals and multi-label trip prediction
- Designed an adaptive update framework using DeepDiff to detect PSL configuration changes, regenerate labels, and incrementally fine-tune models without full retraining
- Automated PSL extraction and mapping through PDF parsing, OCR, and rule-based processing, improving scalability and robustness under evolving operating conditions

GitHub Repository

Real-Time Cryptocurrency Market Analytics Platform | *Python, Apache Kafka, Streamlit, Docker*

- Built a real-time market monitoring platform by ingesting live BTC/USDT trade data from Binance WebSocket feeds
- Engineered streaming pipelines to aggregate high-frequency tick data into OHLCV indicators for trend and volatility analysis
- Created an interactive dashboard for real-time visualization of market activity and trading signals without database polling
- Built low-latency event-driven analytics workflows for real-time market monitoring and trading signal generation

GitHub Repository

StudyHub - Full-Stack Learning Platform | *React, Node.js, Express, MongoDB, Docker*

Preview

- Built a full-stack LMS supporting students, instructors, and admins with role-based access control, course creation, quizzes, and a threaded discussion forum
- Implemented secure authentication with JWT access/refresh token rotation, Google OAuth, email verification, and password reset flows
- Integrated a Gemini-powered AI study assistant embedded per-course, context-aware of lesson content, to answer student questions in real time
- Built automated PDF certificate generation and an XP-based gamification system (streaks, badges, leaderboard) to drive engagement

GitHub Repository

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Probability & Statistics

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, JavaScript

Systems & Tools: Linux, Shell Scripting, Git, Docker, Apache Kafka

Core Areas: Systems Programming, Operating Systems, Computer Networks, Distributed Systems, Multithreading, Production Debugging, Log Analysis, Data Structures & Algorithms

Data & Analytics: Statistical Analysis, Time-Series Data Processing, Machine Learning

Frameworks: PyTorch, TensorFlow, React.js, Next.js

Soft Skills: Problem Solving, Analytical Thinking, Communication, Attention to Detail

LEADERSHIP & EXTRACURRICULAR

Mentor - Product Club, IIT Jodhpur

2025 - Present

- * Mentor students in product management, case-solving, and interview preparation through workshops and one-on-one guidance.

Head - Entrepreneurship Cell, IIT Jodhpur

2024 - 2025

- * Led the planning and execution of entrepreneurship initiatives and coordinated cross-functional teams for campus events.
- * Managed an event during **Prometeo**, IIT Jodhpur's annual techno-management festival, overseeing presentations, participant coordination, and end-to-end event execution.

CERTIFICATIONS

- **CodeClash: The August Arena** – Certificate of Excellence, Foundations Round (C++, Problem Solving) Aug 2025
- **Machine Learning Specialization** – Coursera (Supervised ML, Advanced Learning Algorithms, Unsupervised Learning & Reinforcement Learning) 2023
- **React Certification** – PW Skills Dec 2024
- **Power BI: Beginner to Pro Workshop** – Pragmatic Works Dec 2024
- **Deep Learning Specialization & Agentic AI LLMs** – DeepLearningAI